

Carlos Gael Hernandez Castro

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EDUCATION

University of Texas at El Paso (UTEP)

Bachelor of Science in Electrical Engineering, Minor in Finance. GPA: 4.0/4.0.

El Paso, TX

Expected May 2029

EXPERIENCE

Hardware and AI Data Intern

Jun 2026 – Aug 2026

ArcellAI

San Francisco, CA

- Built a Python data loader for LIBERO robot grasp-prediction tasks using class inheritance, adapting PyTDC's loader architecture and categorizing tasks as single- vs. multi-instance prediction.
- Managed GitHub repository access and structure for the loader project, coordinated onboarding for new team members, and tracked development priorities in ClickUp.

Ground Station Engineer

Jan 2025 – Present

Sun City Rocket Team (UTEP)

El Paso, TX

- Designed a real-time telemetry ground station for flight data acquisition and sensor validation, supporting a 2nd place finish (30k COTS, IREC 2025) and 1st place at the 2025 Lone Star Cup.
- Built a DAQ system interfacing 4-20 mA pressure sensors and K-type thermocouples via Arduino Mega.

FIRE Venture Fellow

Mar 2026 – Jun 2026

Hunt Center for Entrepreneurship, NMSU Arrowhead Center

Las Cruces, NM

- Evaluated a live regional startup (Negawatt Industries), scoring market, team, product, traction, and timing from public data, then modeled seed and Series A dilution, vesting, and cap table structure to assess investment readiness.
- Wrote a full investment thesis on AI-powered grid orchestration (energy/climate tech), covering market opportunity, competitive landscape, and key risks, citing primary drivers like FERC Order 2222 and the \$369B Inflation Reduction Act.

Appliance Technician

Jan 2019 – May 2024

Independent

El Paso, TX

- Repaired 100+ washers and dryers (analog and circuit-based) over 4+ years of service work.
- Resolved 70%+ of electrical and mechanical failures on the first visit.
- Managed 20-30 client service calls per month.

PROJECTS

Multiplexed Custom Solenoid Driver PCB

Aug 2026

Independent Project

- Architected and routed a 2-layer driver board that addresses 48 solenoid channels across 8 Braille cells from a single microcontroller, pairing four decoders with source/sink driver stages for per-dot control, enabling full-array validation before committing to the production device.

Miner Spark Research Project

Sep 2025 – Nov 2025

Researcher, University of Texas at El Paso

- Developed a modular edge vision UAV attachment with on-device facial verification, analyzing lighting effects on recognition accuracy to validate hardware-software integration for low-cost drones.

AWARDS & HONORS

- **BorderHack '26: 1st Place** — Built Wayve, a smart cane for visually impaired navigation, in 24 hours with a 4-person team; recognized for hardware/software integration and accessibility-first design.
- **The Sense Cube: 1st Place at three pitch competitions** — Miner Pitch (60+ entries, \$3,000); Tacos & Tech-ila (\$2,000); Pioneers21 4th Annual (Microsoft-funded, \$1,000). \$6,000 total.
- **SOLACYT International Competition: Top 3 out of 25,000+ students** — Across Ibero-America (Chile, Colombia, Uruguay, Mexico) with a robotic skeleton puppet that improved students' anatomy knowledge by ~60%.
- **Dell Tech Academy Supply Chain Case Competition: 3rd Place** — Competed in Dell Technologies' two-day case competition on global supply chain design and optimization.
- **UTEP Excellence Scholarship Recipient** – April 2025

SKILLS

Hardware: Circuit design, PCB design, Schematic capture, Prototyping, Microcontrollers (ESP32, Raspberry Pi, Arduino)

Programming and Software: Python, C++, MATLAB, Arduino IDE, KiCad, Fusion 360, Autodesk Certified User

Languages: English (Fluent), Spanish (Fluent)